# ArtExtract Task 1 - Full Tuned E8 Results

This notebook summarizes the final full-dataset run from `outputs/full_tuned_e8`.

In [1]:
```
from pathlib import Path
import json
import pandas as pd
import matplotlib.pyplot as plt

OUT = Path('../outputs/full_tuned_e8')
metrics = json.loads((OUT / 'val_metrics.json').read_text())
hist = json.loads((OUT / 'training_history.json').read_text())
metrics
```

Out[1]:
```
{'style': {'accuracy_top1': 0.6233269598470363,
  'macro_precision': 0.3677733908742082,
  'macro_recall': 0.3499724054234531,
  'macro_f1': 0.3489133263826013,
  'weighted_f1': 0.6033269937505233,
  'balanced_accuracy': 0.3499724054234531},
 'artist': {'accuracy_top1': 0.5232632249840663,
  'macro_precision': 0.4838010955214084,
  'macro_recall': 0.4922035992135204,
  'macro_f1': 0.47292502745241644,
  'weighted_f1': 0.5054354954413274,
  'balanced_accuracy': 0.4922035992135204},
 'genre': {'accuracy_top1': 0.6080305927342257,
  'macro_precision': 0.49308354813833377,
  'macro_recall': 0.4722671109396708,
  'macro_f1': 0.468773324078722,
  'weighted_f1': 0.5927981694523236,
  'balanced_accuracy': 0.4722671109396708},
 'artist_top5': 0.8538347142553644,
 'samples_val': 4707}
```

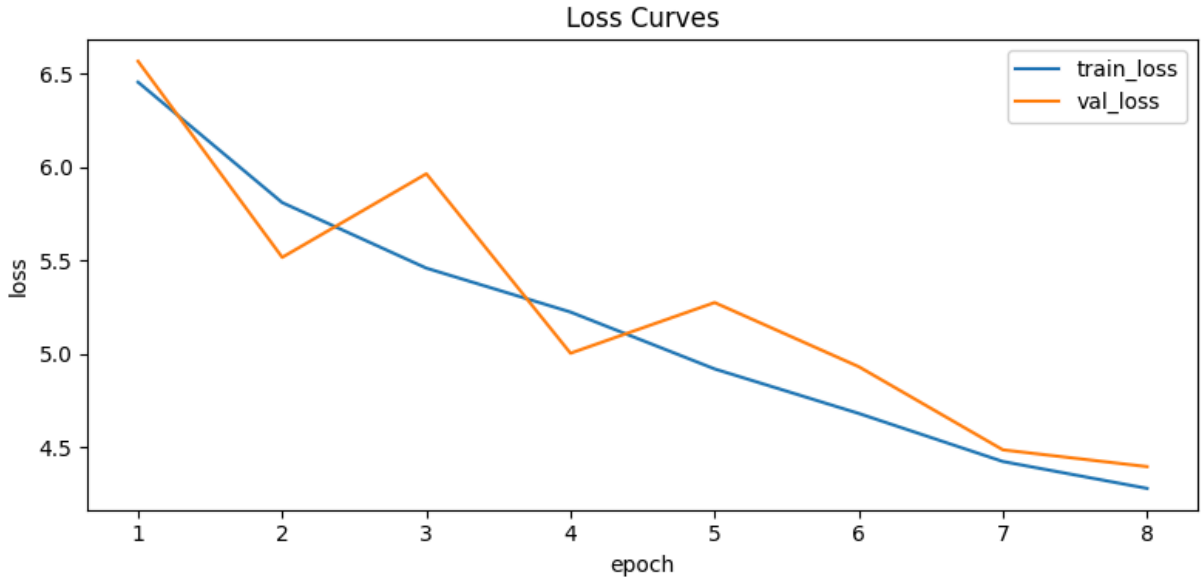
In [2]:
```
summary = pd.DataFrame([
    {'task':'style','top1':metrics['style']['accuracy_top1'],'macro_f1':metr
    {'task':'artist','top1':metrics['artist']['accuracy_top1'],'top5':metric
    {'task':'genre','top1':metrics['genre']['accuracy_top1'],'macro_f1':metr
])
summary
```

Out[2]:

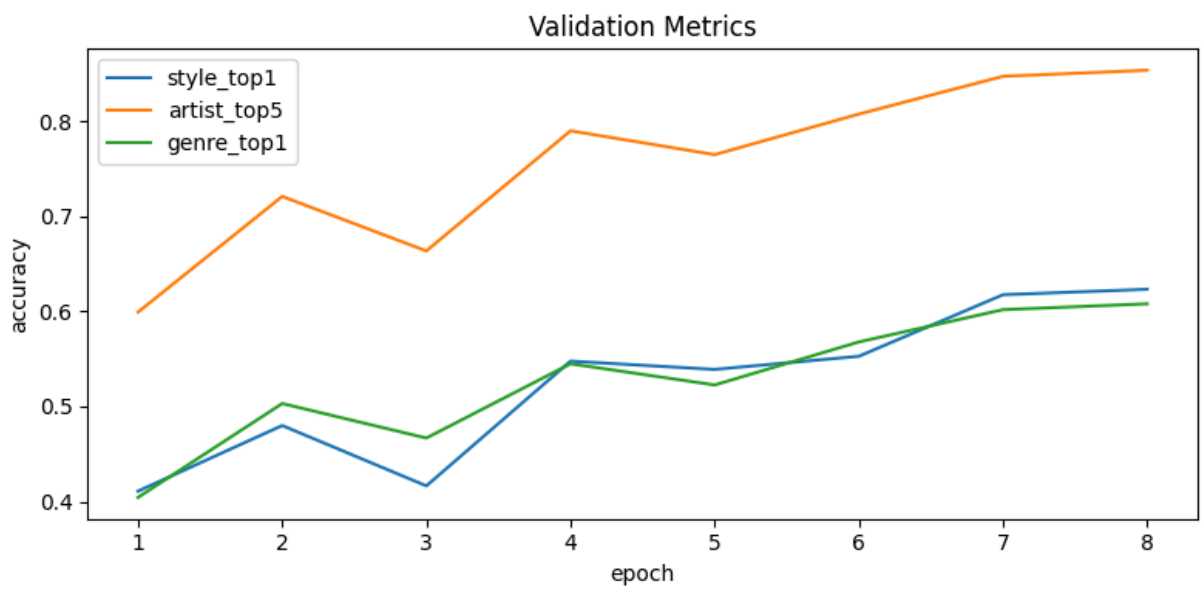
| | task | top1 | macro_f1 | top5 |
|---|---|---|---|---|
| **0** | style | 0.623327 | 0.348913 | NaN |
| **1** | artist | 0.523263 | 0.472925 | 0.853835 |
| **2** | genre | 0.608031 | 0.468773 | NaN |

In [3]:
```
epochs = [r['epoch'] for r in hist]
plt.figure(figsize=(8,4))
```

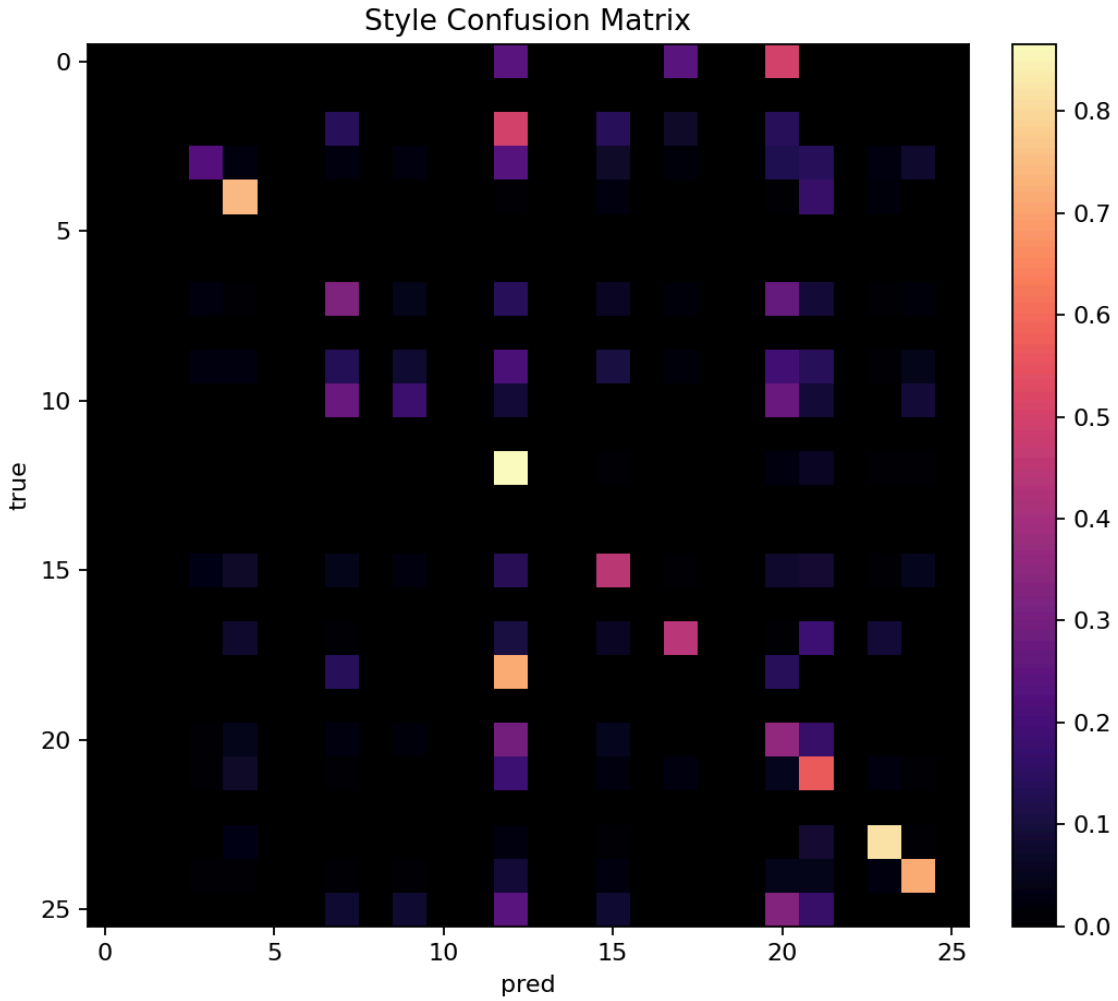
```
plt.plot(epochs, [r['train_loss'] for r in hist], label='train_loss')
plt.plot(epochs, [r['val_loss'] for r in hist], label='val_loss')
plt.xlabel('epoch'); plt.ylabel('loss'); plt.title('Loss Curves'); plt.leger
plt.show()
```

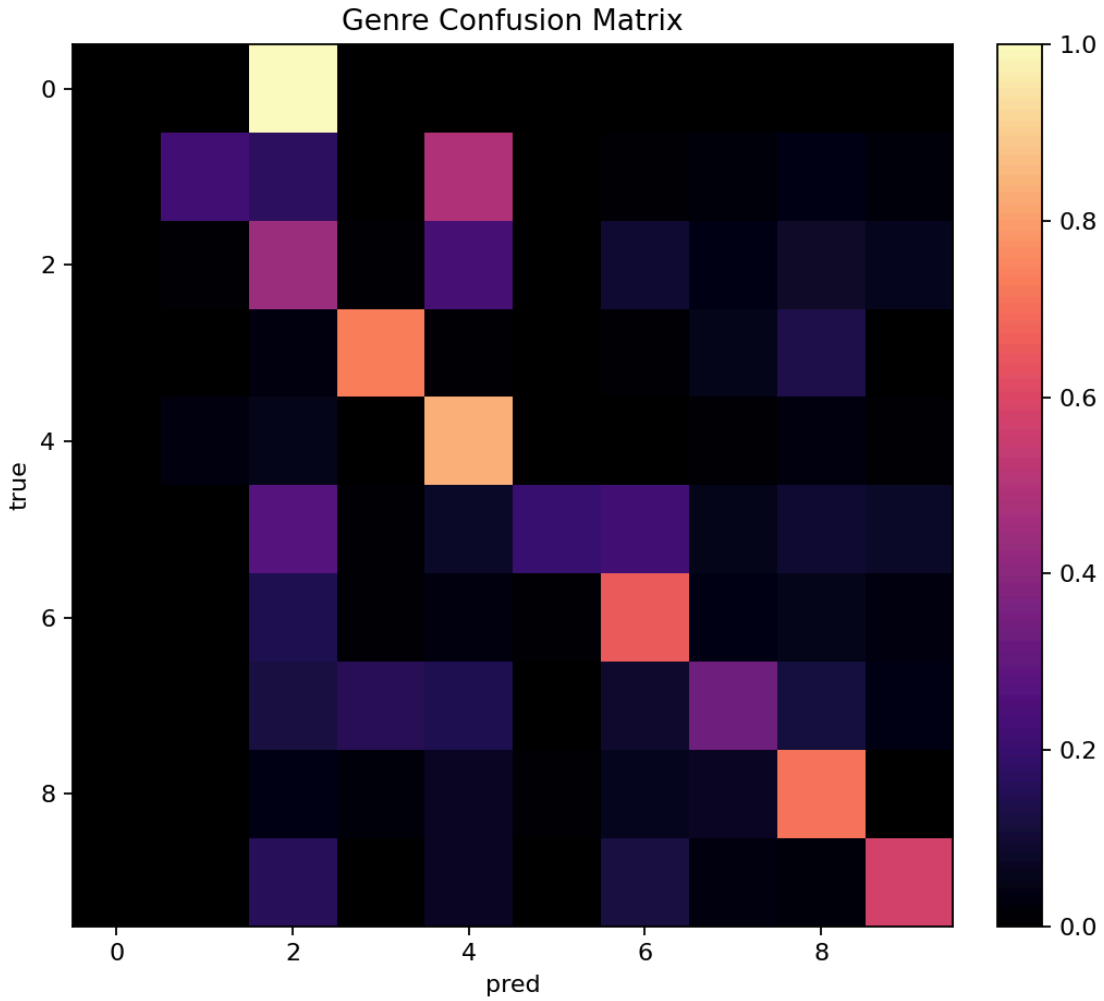


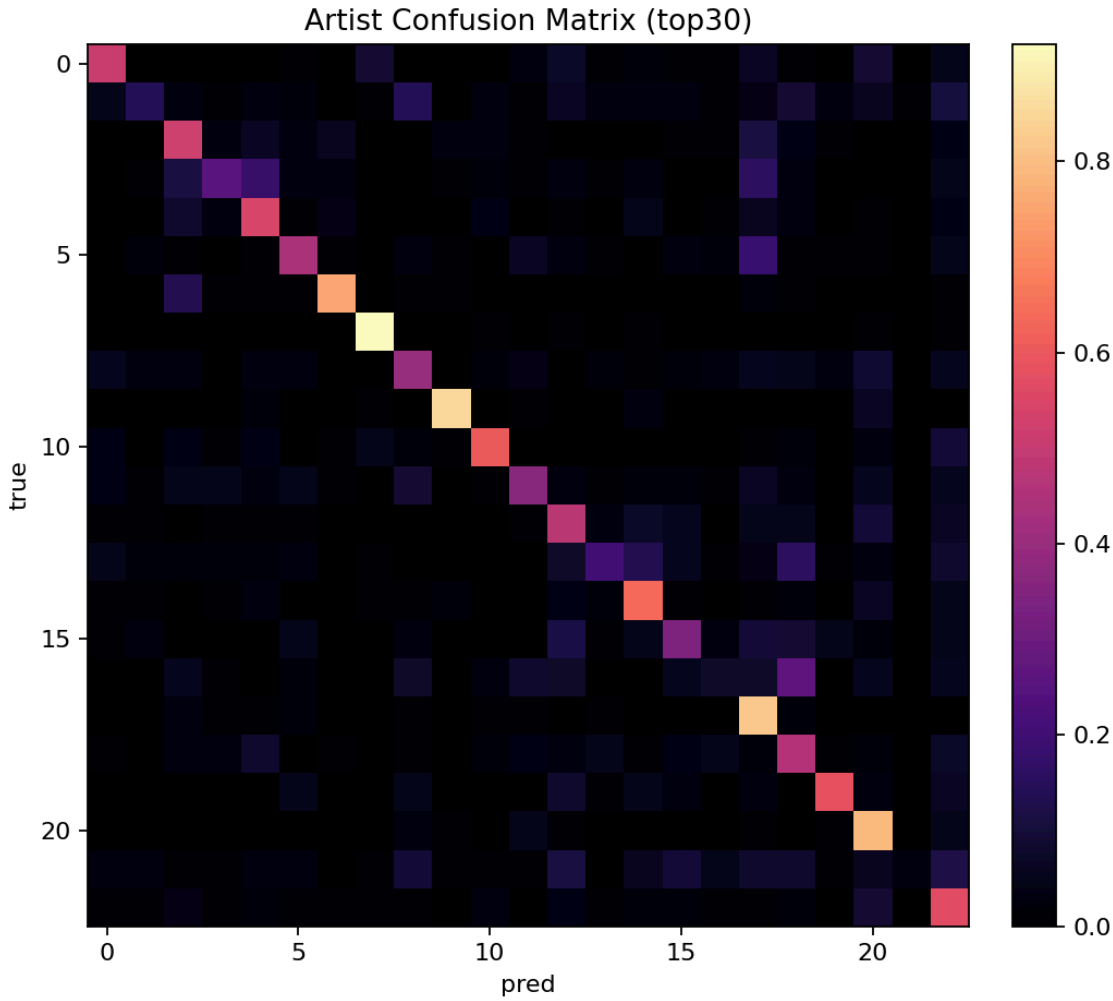

In [4]:
```
plt.figure(figsize=(8,4))
plt.plot(epochs, [r['val_style_top1'] for r in hist], label='style_top1')
plt.plot(epochs, [r['val_artist_top5'] for r in hist], label='artist_top5')
plt.plot(epochs, [r['val_genre_top1'] for r in hist], label='genre_top1')
plt.xlabel('epoch'); plt.ylabel('accuracy'); plt.title('Validation Metrics')
plt.show()
```




In [5]:
```
from IPython.display import Image, display
display(Image(filename=str(OUT / 'style_confusion_matrix.png')))
display(Image(filename=str(OUT / 'genre_confusion_matrix.png')))
display(Image(filename=str(OUT / 'artist_confusion_matrix_top30.png')))
```


Style Confusion Matrix
true
pred
0
5
10
15
20
25
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8


Genre Confusion Matrix
true
pred
0
2
4
6
8
1.0
0.8
0.6
0.4
0.2
0.0




In [6]:
```
outliers = pd.read_csv(OUT / 'style_outliers.csv')
outliers.head(20)
```

Out[6]:

| | index | class_id | score |
|---|---|---|---|
| **0** | 629 | 12 | 5.067897 |
| **1** | 3541 | 24 | 4.859658 |
| **2** | 1726 | 21 | 4.349294 |
| **3** | 141 | 12 | 4.111395 |
| **4** | 1645 | 21 | 4.095620 |
| **5** | 2381 | 21 | 4.027822 |
| **6** | 1435 | 12 | 3.963147 |
| **7** | 331 | 12 | 3.962360 |
| **8** | 3317 | 20 | 3.820931 |
| **9** | 1882 | 21 | 3.723417 |
| **10** | 1259 | 12 | 3.625926 |
| **11** | 2465 | 21 | 3.538754 |
| **12** | 4283 | 7 | 3.485398 |
| **13** | 2821 | 9 | 3.470590 |
| **14** | 2357 | 21 | 3.439830 |
| **15** | 664 | 12 | 3.436456 |
| **16** | 311 | 12 | 3.424903 |
| **17** | 3105 | 20 | 3.388291 |
| **18** | 2398 | 21 | 3.256070 |
| **19** | 751 | 12 | 3.226193 |